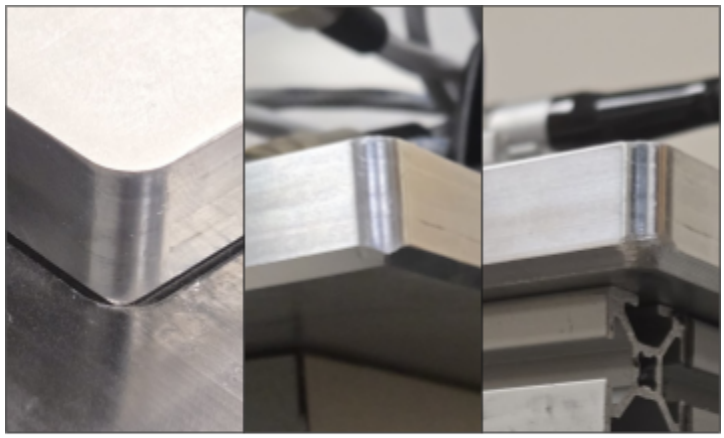
This document shows current obstacles we are facing with the covers and fitment into the ACONITY machine as well as proposed solutions and questions for Dr Downey and Dr Yuan.

1) PROBLEMS
   a) **Corners are not rounded** (already remedied)



(a) (b) (c)

Fig 1.1. (a) The sharp corner not sitting correctly in the chamber seat in the ACONITY, (b) the sharp edge where the chamfers meet, and (c) corner filed down to dimension

   b) **Thickness is 1mm in excess** (MAIN PROBLEM)

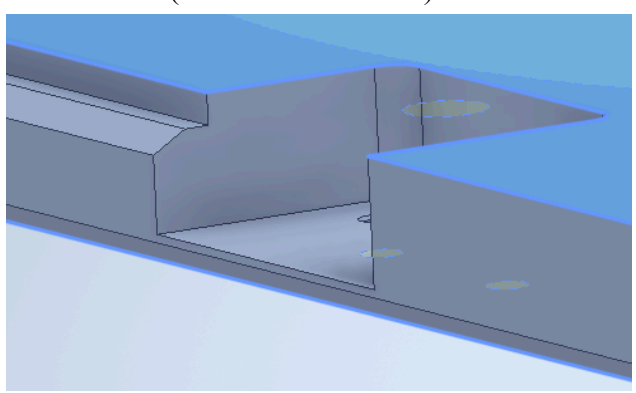


Fig 1.2. Distance between the top and bottom faces is 15.875mm.

   c) **Locking mechanism is lowered -** According to Dr. Yuan this should not affect a cover of the correct dimensions, so addressing the previous problems is the priority.

2) MACHINING

a) Option 1 - mill down <u>side rails and magnet seats</u> with the reasoning that the bottom face might not be able to go deeper into the chamber.

b) Option 2 - mill down the <u>bottom chamfers</u> in the hopes that the cover will sit deeper into the recess and effectively lower the top surfaces (the side rails where the locking jaws clamp and the magnet seats).

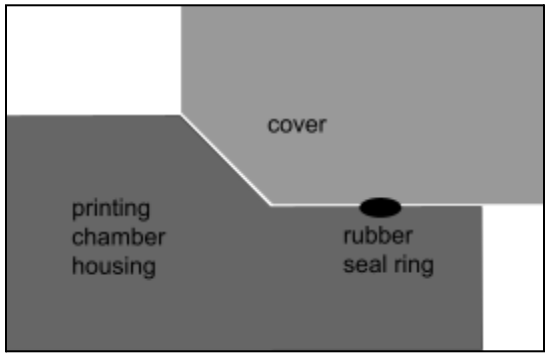


Fig 2.1. Increasing the chamfer depth will not allow the cover to sit lower, because the bottom face is always in contact with the seat (surface with seal ring).

Dr.Yuan initially was against machining the entire bottom face, and increasing the bottom chamfers will not solve the problem. We must continue with Dr. Yuan's original idea (Option 1).

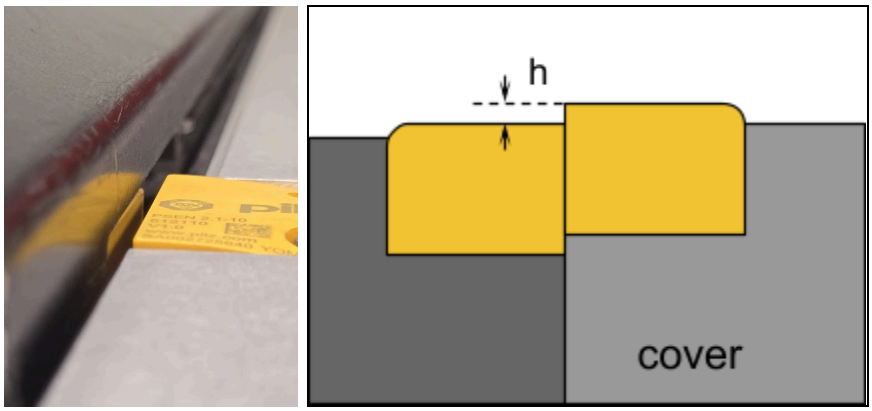


Fig 2.2. (Left: photo of test fit, Right: diagram) The magnet on the cover appears to be more than 1mm off in alignment tests on the cover. We will determine the value of *h* and correct the model.

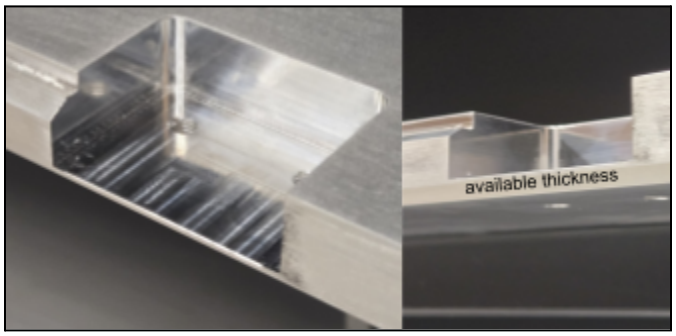


Fig 2.3. If we machine down the magnet seat, there will still be plenty of material left. The available thickness is more than *h*.

3) HARDWARE

a) Goal: Make the scanning hardware able to fit in situ and not obstruct the printing process. (As requested by Dr. Downey)

b) Solution: Move the galvo out of the way of the welding laser and center cylinders:

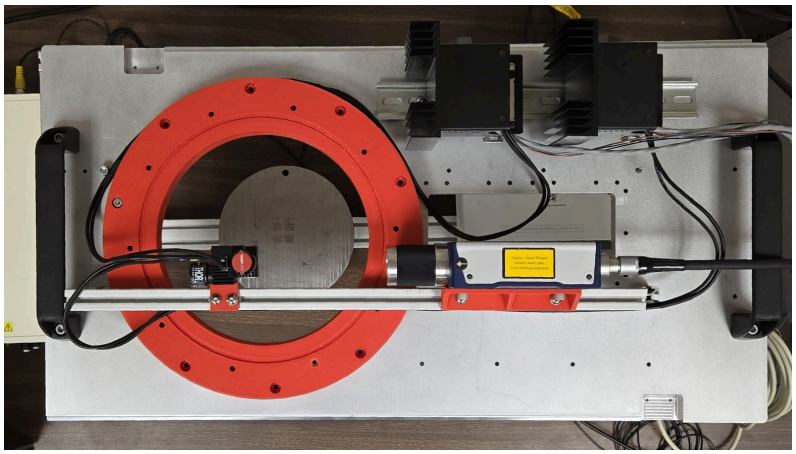


Fig 3.1. Current configuration. The telescoping center cylinders cannot be installed without hitting the galvo.

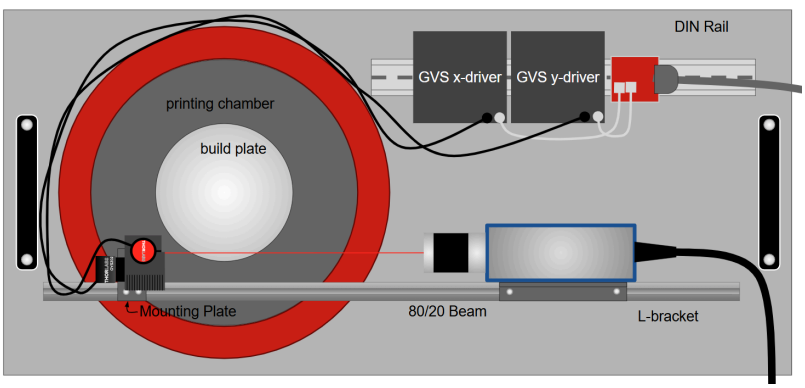


Fig 3.2. Proposed orientation so that the center cylinder can be installed and so that the welding laser is given a wide berth. (We just slide the galvo back, and reprint the cylinder housing so that the galvo can sit inside of a recess and create clearance for the center cylinders.